AI/ML INTERNSHIP · DEVELOPERSHUB CORP · TASK 02

AAPL Stock *Prediction*

Apple Inc. · Jan 2020 → Apr 2026 · Linear Regression + Random Forest · 1,561 Trading Days

02

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LIVE TEST ERROR	BEST R^2 SCORE	AVERAGE DAILY ERROR	WINNER
1.41%	0.9428	$4.48	Linear Reg.
$269.91 predicted · $266.17 real	94.3% of price movement explained	On a ~$220 stock = 2% avg error	AAPL trends smoothly → linear wins

MODEL EVALUATION · TEST SET · 313 DAYS

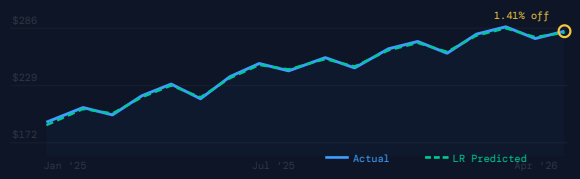
Performance Comparison

	LINEAR REGRESSION 🏆 WINNER	RANDOM FOREST
MAE	$4.48	$10.77
RMSE	$6.41	$13.83
R^2	0.943	0.734

INPUT DATE	ACTUAL CLOSE	LR PREDICTED	ERROR
Apr 20 '26	$266.17	$269.91	1.41%

PLOT — ACTUAL VS PREDICTED · TEST PERIOD

AAPL Jan 2025 → Apr 2026



1,561	1,248	313	15
TOTAL DAYS	TRAIN (80%)	TEST (20%)	FEATURES

KEY FINDINGS — PLAIN ENGLISH

What This Actually Tells Us

01 Yesterday's price is almost the whole story

The single strongest predictor is simply what Apple closed at yesterday. All 15 features help, but `Prev_Close` alone explains most of the R^2=0.94. Apple doesn't teleport — it moves gradually.

02 Linear beat Random Forest because AAPL is boring

Random Forest is designed for chaotic, spiky data. Apple trends smoothly and predictably. Using a complex model on simple data adds noise — like using a sledgehammer to crack an egg.

03 Moving averages show the model where the wind is blowing

Without MA5–MA21, the model only knows yesterday's price. With them, it knows the direction and speed of the trend over the past week and month. This is what pushed R^2 above 0.90.

04 94.3% on data the model never saw = real learning

Trained on 2020–2025. Tested on 2025–2026 — a period it never studied. Scoring 0.94 R^2 on completely unseen data means the model genuinely learned patterns, not just memorized examples.

DEBUGGING JOURNEY · 6 ATTEMPTS

Road to R^2=0.94

Try 1	-0.16	MultiIndex column bug
Try 2	-0.26	Feature data leakage
Try 3	0.19	Too little training data
Try 4	-12.6	NVDA stock split in data
Try 5	0.84	Switched to AAPL, clean
Final	0.943	Full features + fixed ✅

DATA LEAK AUDIT ✅

No Leakage Found

- ✅ `All features use .shift(1)` — Only yesterday's data fed as input — never today's
- ✅ `Rolling averages shifted before rolling` — `.shift(1).rolling(n)` — excludes today from every MA
- ✅ `Scaler fit only on train data` — `fit_transform` on train, `transform` only on test
- ✅ `No shuffle on time series split` — Chronological — past trains, future tests
- ⚠ `Raw OHLCV columns still in data` — Not in FEATURES list → not passed to model. No actual leak.

15 FEATURES — PAST DATA ONLY

What the Model Sees

Prev_Close/H/L/O	yesterday OHLC
Prev_Volume	yesterday volume
MA5 / MA7	short-term trend
MA14 / MA21	medium-term trend
Price_Change	1-day momentum
Price_Change5	5-day momentum
Daily_Range	High - Low
Volatility5	5-day std dev
Volume_MA5 / Change	volume activity

TECH STACK

Libraries & Skills

- `yfinance` — Live Yahoo Finance data
- `pandas · numpy` — Data engineering
- `scikit-learn` — Models + metrics + scaler
- `matplotlib · seaborn` — Visualization

Time Series · Feature Eng. · Regression · MAE·RMSE·R^2 · No Data Leak · Live Validation